M T W T F S S
Page No.:
Date:
YOUVA

MC :- Assignment 1

1] What is Antenna & Explain different types of Antenna.
2] Compare all Mobile Generation ie, 1G, 2G, 3G, 4G & 5G.
3] Describe GSM in detail.
4] What are the characteristics of SIM.
5] Explain Mobile Originated Call & Mobile Terminated call.
6] What are diff. types of Handover in GSM.

UTI Imp told = Mobile IP
Difference
Hidden & exposed
Assignment Q No.
Mobile terminated call q no.
GSM architecture and all Architecture.

Note :- Max 2 pg ans required

M T W T F S S
Page No.:
Date:
YOUVA

MC :- UTI QB 2

1] What is an Antenna & Explain different types of Antenna.
2] Compare all Mobile Generation ie., 1G, 2G, 3G, 4G & 5G.
3] Describe GSM in detail
4] Explain SIM.
5] Explain How Mobile Originated call work.
6] What are different types of Handover in GSM.
7] Explain Mobile IP.
8] Explain Hidden station & exposed station problem in WLAN.
9] Comparison between DSSS & FHSS.

GSM.
Hidden
Mobile IP.
SIM.

12/2/26

* Answers for UTI QB :-

Q] Compare :- DSSS & FHSS

Parameter	DSSS	FHSS
Abbreviation	Direct Sequence Spread Spectrum.	Frequency Hopping Spread Spectrum.
Concept	DSSS spread signal continuously across a wide band by multiplying data with a "chip code".	FHSS rapidly "hops" the carrier frequency between multiple narrow channels.
Interference Immunity	Low	High
Receiver complexity	More complex	Less complex
Bandwidth utilization	Always uses total bandwidth ~~at once~~.	Uses only a portion of total bandwidth at a time.
Inshort	DSSS spread signal	FHSS jump signal
Power Consumption	High	Low
Eg.,	CDMA Wifi	Bluetooth
Security	Secure bcz of Secret code	Secure bcz of hopping pattern

Q] Compare all Mobile Generation i.e., 1G, 2G, 3G, 4G, 5G.

Feature	1G	2G	3G	4G	5G
Introduce	1984	1991	2000s	2010s	2020s
Technology	Amps TACS	D-Amps GSM	W-CDMA	LTE Wimax	LTE Advanced MIMO, mmWave
Multiplexing	FDMA	TDMA/CDMA	CDMA	CDMA	CDMA
Signal	Analog (voice only)	Digital (GSM/CDMA)	Digital Broadband	IP based	IP-based
Switching type	Circuit Switching	Circuit Switching	Packet Switching	Packet Switching	Packet Switching
Speed	2.4 kbps	14.4 Kbps	1.4 Mbps	100 Mbps	>10 Gbps
Bandwidth	Analog	25 MHz	25 MHz	100 MHz	6GHz
Carrier freq.	30 KHz	200 kHz	5 MHz	15 MHz	28 GHz
Handover	NA	Horizontal	Horizontal	Horizontal/Vertical	Horizontal/Vertical
Services	Voice calls only	Voice + SMS	Video calls Web	High speed Video, Gaming	IOT, AR/VR Ultra low latency

What is Antenna & Explain the different types of Antenna.

• Antenna

1) An Antenna is a device that converts Electromagnetic radiation in space into Electrical current in conductor or viceversa.

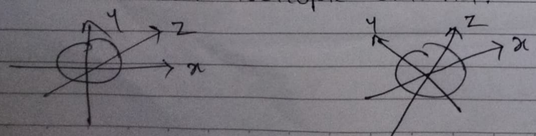
2) The radiation pattern of an antenna describe the relative strength of the radiated field in various direction from the antenna at a constant distance.

3) In reality the radiation pattern is 3-dimensional but usually the measured radiation patterns are a 2-dimensional slice of the 3-dimensional pattern.

4) There are various types of Antenna dissucussed below:

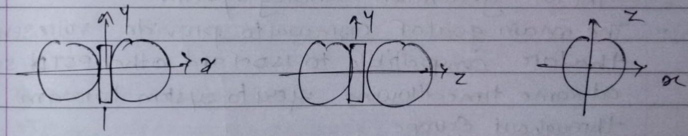
I] Isotropic Antenna: —

- An Isotropic Antenna is an "theoretical" antenna that radiates its power uniformly in all directions.
- In other words, theoretical asotrophic antenna has a perfect "360 degrees" spehenical radiation pattern.
- It is an ideal antenna which radiates equally in all direction & has again of 1 (0dB) i.e., "zero gain and zero loss."
- It is used to compare the power level of a given antenna to the theoretical isotropic antenna.
-



II] Omnidirectional Antenna :-

- Unlike Isotropic antenna, dipole antennas are real antennas.
- The dipole radiation pattern is "360 degrees" in horizontal plane & approx. "75° degree" in vertical plane.
- It is also called "non-directional" antenna because it does not favour any particular direction.
- Dipole antenna are said to behave a gain of 2.14 dB which is in comparision of Isotropic antenna.
- The higher the gain of the Antenna, the smaller the vertical beam width is.
- This types of antenna is usefull in broadcasting a signals to all points of a compass or when signals from all points.



sideview sideview Topview.

Q] Describe GSM in detail.

→

- Global System for Mobile Communication (GSM) is the most successful digital mobile telecommunication system in the world today.
- It is used by over 1000 million people in more than 190 countries.
- Global system for Mobile Communication (GSM) is a globally accepted standard for digital cellular communication. It is an ETSI standard for 2G pan-European digital cellular system with international roaming.
- The basic version of GSM (ie. GSM 900) was founded in 1982.
- GSM is second generation 2G system, replacing the 1st generation analog system.
- The main goal of GSM was to provide voice service that are compatible to ISDN and other PSTN system at same time allowing users to system to roam throughout Europe
- GSM comprises of 3 things :-
  - Radio Subsystem [RSS]
  - Network and Switching Subsystem [NSS]
  - Operation Subsytem [OSS]
- Benefits :-

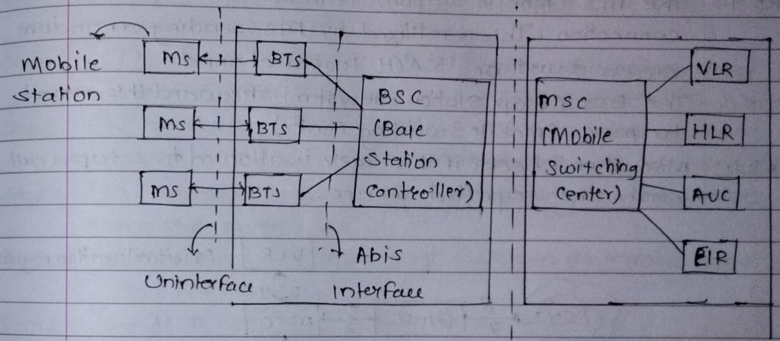
i) Enhanced Security compare to prev. technologies
ii) Voice calling [enhanced one]
iii) International Roaming Services
iv) Textmessages
v) Basic Mobile Internet Service

TDMA = Time division Multiple Access

- Facts about GSM :-

i) Uses TDMA for Signal transmission.
ii) Core concept of GSM was introduced in Bell Laboratries of 1970s
iii) GSM vs CDMA
↳ कोई भी SIM → एक company Mobile, वो ही SIM.

-



1) Mobile station [MS]
2) Base Transceiver station [BTS]
3) Base station Controller [BSC]
4) Mobile switching center [MSC]
5) Home Location Register [HLR]
6] Visitor Location Register [VLR]
7] Authentication center [AUC]
8] Equipment Identity Register [EIR]

Q] Explain Mobile Terminated Call (MTC) & Mobile Originated Call (MOC).

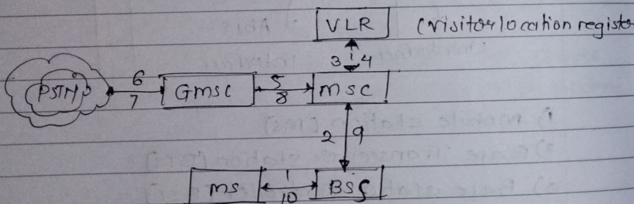
→ I] Mobile Originated ~~Moc~~ call (MOC) :-

It is much simpler to perform a Mobile Originated Call [MOC] compared to MTC. This follows the following steps:

p 1: The MS (Mobile station) transmits the request for a new connection. This is realized by MS sending a random access burst on RACH logical channel.

2: The BSS (Base Station subsystem) forward this request to MSC (Mobile Switching Center).

3,4: MSC then checks if this user is allowed to setup a call with the requested service.



p5-8: The MSC checks the availability of resources through GSM network and into PSTN

p 9,10: If all resources are available ~~of~~; the MSC sets up connection between MS and fixed network.

II] Mobile Terminated Call (MTC):-

This is the situation where a terminal from a fixed network calls a Mobile Station. This involves the following steps:

Step 1: A PSTN user dials the phone no. of a GSM subscriber. The fixed network (PSTN) notices that the number is of the GSM network and forward call setup to Gateway MSC [GMSC].

Step 2: GMSC identifies the HLR for the subscriber and signals the call setup to the HLR [Home Location Register].

Step 3: The HLR now checks whether the number exists and whether the user has subscribed to the requested services.

Step 4: HLR request Mobile subscriber roaming number (MSRN) from the current VLR.

Step 5: HLR receives MSRN. and HLR can determine responsible MSC for the MS.

Step 6: The HLR forward the info. to GMSC.

Step 7: The GMSC Forwards the call setup request to MSC.

Step 8,9: The MSC first request the current status of the MS from the VLR.

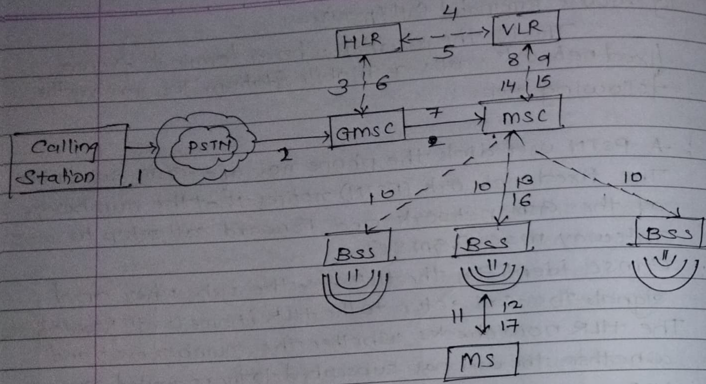
Step 10: If the MS is available the MSC initiates paging in all cells.

Step 11: The BTS of all BSSs transmits this paging signal to the MS

Step 12,13: The MS Answers.

Step 14,15: The VLR does Security Check

Step 16,17: The connection is setup.



Q.47 What are the characteristics of SIM.

→ - If you mean SIM (Subscriber identity Module) in Mobile telecom, here are its key characteristics in simple terms:

1) Unique identification :- you can each sim has a unique IMSI (International Mobile Subscriber identity)

2) Authentication & Security :- Stores secret keys used to authenticate the user with the Mobile networks. Enabled encrypted communication helping prevent fraud and eavesdropping.

3) Subscriber Information Storage :-
- Holds data Like:
- Phone number
- Network authorization detail
- Sometimes contacts and sms.

4) Small Standardization form :- Comes in different size standard ge sms, sim, Micro-sim, Nano sim. and now ESIM.

5) Operator - Specific :- Issued and managed by Mobile network operator, not the phone manufacturer.

3.3 Mobile IP

3.3.1 Mobile IP : Basic Concept

- Mobile IP (or IP mobility) is a communication protocol developed by Internet Engineering Task Force (IETF) standard.
- In mobile IP, nodes continue to receive packets independent of their location which is achieved by modifying the standard IP in a certain way. It is designed such that mobile users can move from one network to another while maintaining a permanent IP address.

3.3.1(a) Need for Mobile IP

- To understand the need for Mobile IP, let us first understand the problem with the internet protocol (IP).
- In the standard IP, a host's IP address is made up of a network identifier and a host identifier. This network identifier specifies the network the host is attached to.
- A host sends an IP Packet with the header containing a destination address, made up of its network identifier and destination identifier.
- As long as the receiver remains connected to its original network, it can receive packets.
- Now suppose the receiver disconnects itself from its original network and joins another network, the receiver would never receive any packets. This is because, the IP address of the host is now topologically not correct in the new network.
- Hence, a host needs a so-called topologically correct address and Mobile IP standard was developed.

3.3.1(b) Goals/Requirements of Mobile IP

MU – Dec. 18

Q. What is the goal of Mobile IP?	**(Dec. 18, 5 Marks)**

1. **Flawless Device Mobility Using Existing Device Address**

   Mobile devices can continue to use their existing IP address even while changing their acti network.

2. **No additional Addressing or Routing Requirements**
   - The same overall scheme for addressing and routing must be maintained as in regular IP. The owner of each device must assign IP addresses in the usual way.
   - New routing requirements must not be placed on the internetwork, like host-specific routes.
3. **Interoperability**

   Mobile IP devices can continue to communicate with other IP devices that have no idea about how Mobile IP works, and vice-versa.

4. **Transparency of Layers**
   - All changes made by Mobile IP must remain confined to the network layer.
   - Other layers like the transport layer and applications must be able to function in the same way as regular IPv4.
5. **Restraining Hardware Changes**
   - A few changes are required to the routers that are used, by the mobile device and the mobile device software for Mobile IP.
   - These changes must be kept to a minimum. Other devices, however, like routers between the ones on the home and visited networks, do not need changes.
6. **Scalability**
   - Mobile IP must allow any device to change from one network to another network, and this must be supported for an arbitrary number of devices.
   - The scope of the connection change must be global. For example, you can use your laptop from an office in London and also use it if you move to Mumbai.
7. **Security**

   Mobile IP must include authentication procedures to prevent unauthorized devices from causing accessing the network and thereby causing problems.

3.3.1(c) Basic Terminology

MU – May 12, Dec. 16

Q. List the entities of mobile IP and describe data transfer from a mobile node to a fixed node and vice versa. **(May 12, Dec. 16, 10 Marks)**